

ENGR 2303 Electrical Science

Review Material for Test 2 and Sample Test

Given Electric circuit diagram, you should be able to find the current, voltage, and power at any element in the circuit using any of the following techniques:

- | | |
|---------------------------|----------------------|
| 1) Source Transformation | 2) Superposition |
| 2) Thevenin's Theorem | 4) Norton's Theorem |
| 5) Maximum Power Transfer | 6) Wheatstone Bridge |

Question 1

Find I_o in the circuit in **Fig. 1** using superposition.

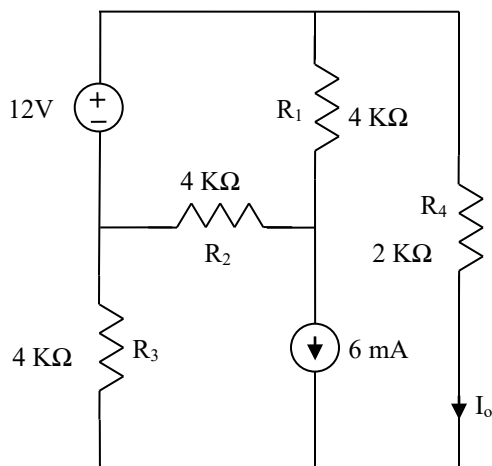


Fig. 1

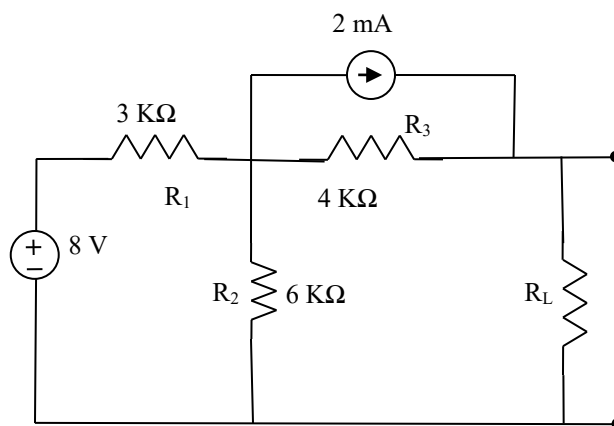


Fig.

Question 2

Find I_o in the circuit in **Fig. 1** using Norton's Theorem.

Question 3

Find R_L for maximum power transfer and the maximum power that can be transferred to the load in **Fig. 2** above.

Question 4

The Wheatstone bridge circuit shown below is used to measure the resistance of a strain gauge. The bridge is balanced when the adjustable resistor is 21.3 ohms. Find the resistance of the strain gauge when the bridge is balanced?

